
Compass — Press Kit

For journalists, podcasters, hackathon judges, and anyone writing about Compass. Permission granted to reuse anything in this file with attribution to “Compass — built solo by Stephen Sookra for the oG APAC Hackathon”.

One-liner (8 words)

A private eligibility firewall for vulnerable workers.

Three-line description

Compass lets a vulnerable migrant worker prove she qualifies for a free service — legal help, shelter, hospital care — without disclosing her name, HKID, employer, or any document field. The service learns only a bucketed 15-minute timestamp and a cryptographic commitment. Built end-to-end on oG: Chain (atomic single-use grants), Storage (encrypted vault), and TeeML / Phala dstack TDX (sealed attestation).

Long description (200 words)

Hong Kong is home to 368,000 Foreign Domestic Helpers — 9.6 % of the local workforce (HK LegCo Research Office, 2025). 17 % are in forced labour and 60 % are deterred from filing Labour Tribunal claims by deportation fear (Justice Centre HK 2016; Mission for Migrant Workers 2023). Across APAC the migrant-worker population is 27.2 million (ILO, 2024). The legal and humanitarian services that exist to help them require disclosure of identifying data: name, HKID, employer, visa status. Those disclosures are exactly the levers that abusive employers and traffickers use to find, retaliate against, and silence the workers seeking help. The status quo asks workers to choose between getting help and being safe.

Compass eliminates the disclosure. It is a privacy-preserving eligibility firewall: a vulnerable worker carries an SD-JWT verifiable credential encrypted in her browser vault (AES-256-GCM with a non-extractable key in IndexedDB); a sealed-inference receipt-signer running inside a Phala dstack TDX trusted execution environment evaluates an eligibility policy against selectively-disclosed claims; the result is committed to oG Chain as a `ReceiptIssued` event whose fields are non-identifying — only a 15-minute timestamp bucket, a policy ID, a nullifier, and a cryptographic commitment to the agent’s on-chain identity. A subpoena reaches the bucketed timestamp and the commitment. Nothing else exists to be disclosed.

What's built

- oG Aristotle mainnet (chainId 16661): AgentRegistry + CompassHub contracts, three policies registered.
- oG Galileo testnet (chainId 16602): parallel demo path for judging.
- Phala dstack TDX receipt-signer with per-receipt RA quote binding `(signer, image, receiptId)`.
- AES-256-GCM browser vault using non-extractable WebCrypto keys.
- SD-JWT VC issuance + verification end-to-end.
- 5-language kiosk mode (English, Filipino, Bahasa Indonesia, Bahasa Malaysia, Cantonese).
- 3D audit-log visualization at `/audit-graph.html`.

Business Impact

TAM — 368,000 FDHs in Hong Kong (LegCo 2025) · 27.2 million migrant workers across APAC (ILO 2024) · US\$1.087 billion in personal remittances out of Hong Kong annually (World Bank 2024).

Cost per incident when an FDH is deported because identity disclosure at intake was subpoena-reachable:

- Worker side: ≈ **US\$22,200** lost (HK\$152K remaining contract wages + HK\$21K outstanding recruitment debt; HK Labour Dept MAW Sept 2025 + Amnesty 2013).
- HK government side: ≈ **US\$14,100** lost (≈ HK\$71,500 LAD civil application + HK\$40–60K deportation processing; LAD FY2024/25 statistics + HK Budget Head 70).

Sustainability — open-source. Free for migrant-worker NGOs. AGPL core + commercial dual-license for non-NGO deployments. Phala dstack TDX hosting ≈ US\$15/year per receipt-signer instance. 12-month grant ladder: Phala Builders Program (\$10–50K) → oG ecosystem (\$10–100K) → EF PSE + Mozilla Technology Fund (\$30–250K) → Open Society Migration Initiative + Luminate + HK Jockey Club Special Projects (\$150K–1M+). Target by month 18: ≈ US\$60K/year recurring (managed hosting + dual-license).

Full sources in `docs/whitepaper.md` → §Business Impact.

Founder quote

“The thing that wouldn’t let me sleep was the asymmetry. A woman escaping wage theft has to hand her HKID to the same legal-aid intake that an employer’s lawyer can later subpoena. The technology to eliminate that asymmetry already exists in scattered pieces — SD-JWT selective disclosure, sealed inference, on-chain commitments. Compass just wires them together in the order that matters.”

— Stephen Sookra, builder

Honest limits

We are explicit about what Compass does *not* do. The full list lives at `docs/honest-limits.md`. Headline items:

- Compass does not defeat coercion. An abusive employer who can see the worker’s screen at the moment of disclosure can still read the plaintext SD-JWT VC before encryption.
- 15-minute timestamp buckets are not full k-anonymity. With enough side-information, statistical re-identification of edge-case workers may be possible. We default to coarse buckets and document the trade-off.
- SD-JWT VC standardization is still draft. The current implementation pins to `draft-ietf-oauth-sd-jwt-vc-15`; we will roll forward as the standard lands.

How to cite

```
@misc{compass2026,  
  title = {Compass: A Private Eligibility Firewall on OG},  
  author = {Sookra, Stephen},  
  year = {2026},  
  url = {https://github.com/StephenSook/Compass-OG-},  
  note = {OG APAC Hackathon Track 5 (Privacy \& Sovereign Infrastructure)}  
}
```

Logos and screenshots

Asset	Path	Use
Compass favicon	<code>app/src/app/favicon.ico</code>	favicons, social cards
Architecture diagram	<code>app/src/components/about/ArchitectureDiagram.tsx</code> (rendered on <code>/about</code>)	technical articles
Subpoena scene screenshot	live at <code>https://app-psi-pied.vercel.app/clinic/subpoena</code>	“what subpoena reveals” graphic
3D audit view	live at <code>https://app-psi-pied.vercel.app/audit-graph.html</code>	visualization in coverage
Whitepaper PDF	<code>docs/whitepaper.pdf</code>	embedded reference

We do not have a dedicated wordmark or brand color yet — please use plain “Compass” in the body type of your publication. If you need a visual asset and the above does not suit, email stephensookra@gmail.com.

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- Live frontend: <https://app-psi-pied.vercel.app>